Enzyme analysis: Enzyme activity, enzyme assay, and enzyme kinetics

Enzyme activity

Enzyme activity is the rate of enzyme reaction—generally expressed as units of substrate converted (or product formed) per time unit. Enzyme activity depends on numerous factors. Therefore, measuring enzyme activity is a precise undertaking that requires technical knowledge, skill, and technique, as well as modern laboratory instruments.

Enzyme assays

Enzyme assays are laboratory methods that measure the rate of enzyme reactions (enzyme activity). These assays also help to develop analysis technologies and to study enzyme catalytic mechanisms. They are vital for the study of enzyme kinetics and enzyme inhibition. Usually the assay is carried out by determining the enzyme activity with and without activation by an added coenzyme. The activity can be monitored by measuring changes in concentration of substrates or products during the reaction.

Enzyme kinetics

Enzyme kinetics is the study of the chemical reactions that are catalyzed by enzymes. In enzyme kinetics, technicians measure reaction rates and investigate the effects of the varying conditions of the reaction. Studying an enzyme's kinetics can reveal the catalytic mechanism of this enzyme, its role in metabolism, how its activity is controlled, and how a drug or an agonist might inhibit the enzyme.

Enzyme assay methods: Many variables. Precision required.

The enzymatic assay can be direct, indirect, or coupled. In the case of direct assay, substrate is added to the sample, the end-product is determined, and the substrate or assay reagent is directly modified by the enzyme. The signal is generated by interaction with another reagent or another reaction.

Developing a reliable analytical method for enzyme assay or enzyme activity involves many different steps. Overall method development is tedious and time consuming. Method development starts with identifying the key method variables through design of experiment (DoE), which requires many sets of samples to be tested for enzyme activities. Flexible method parameters for each enzyme type: measuring wavelength, blank measurement, buffer addition, reagents additions, substrate addition, enzyme specific incubation temperature, enzyme specific

incubation time and data collection duration all make the enzyme assay method development and method transfer much more simple and reliable from research and development (R&D) to QA/QC labs.

Depending on the type of enzymes and their sensitivity, many different analytical methods are used. Most enzyme assays are based on spectroscopic techniques, with the two most used being absorption and fluorescence.

Enzyme assays based on photometry, fluorometry, 96-, 384-, or even 1536-well format microplate offer a high-throughput alternative to the traditional spectrophotometers. The microplate format is convenient for high-throughput analysis using a 200μL assay volume and is commonly used in life science applications. However, the microplate method suffers from temperature stabilization, absorption correction, and edge effect.